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Paper

A Decentralized Voltage Regulation Scheme Using Improved Volt-Var Function of PV Smart Inverter

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With the growing distributed PV installation rate in distribution systems, voltage regulation difficulties such as local voltage violations and fluctuations have become common. To solve the voltage regulation problems, the local voltage regulation method using volt-var (VV) function is effective for its high regulation speed, high accuracy, and flexibility. However, there are still hurdles on real application, such as parameter setting difficulties, insufficient voltage fluctuation mitigation effect, and unfairness of reactive power generation between PV customers. To further improve the VV function, this paper proposes a PID closed-loop based VV function and mode-switching function of PV smart inverter (SI). To validate the proposed methods, numerical simulations were conducted using a 6-feeder distribution system model based on the JST-CREST 126 distribution feeder model. According to the simulation result, the proposed method can better mitigate the local voltage violation compared to basic VV function based on the VV curve, also improve the fairness between PV customers, while the total reactive power generation and tap operations increases for the aim of more adequate voltage regulation. © 2024 The Authors. *IEEJ Transactions on Electrical and Electronic Engineering* published by Institute of Electrical Engineers of Japan and Wiley Periodicals LLC

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1. Introduction

To control the rate of global warming, the development of renewable energy, especially photovoltaic (PV) generation, has been promoted globally. After the implementation of feed-in tariff policy [1], there has been rapid growth in the PV installation capacity in Japan since 2012. Distributed PV generators brings various challenges to voltage regulation in distribution systems.

The high penetration rate of distributed PV generation in distribution systems can cause local voltage violation and fast voltage fluctuations [2,3]. The on-load tap changer (OLTC) is one of the most widely used voltage regulation device in current distribution systems. OLTCs adjust the transformer ratio by changing the tap position and regulates the voltage in a wide area. The adjusting range per tap is ∼100 V in 6.6 kV medium-voltage system, and the regulation speed is limited due to the mechanical action [4]. Meanwhile, as power flow becomes more complicated in low-voltage systems due to distributed PV generation, it is possible that the voltage in medium-voltage (MV) system is well regulated, while local voltage violation remains in low-voltage system. Therefore, decentralized voltage regulation is now essential.

To realize distributed voltage regulation, the PV smart inverter (SI) has been promoted by the Electric Power Research Institute (EPRI) in the United States. It can adjust the active power and reactive power injection from PV generators to the power system [5]. As the control is based on inverter circuit and local sensor measurement, the responding speed is fast. Also, PV SIs can communicate with distributed energy resources management systems (DERMS), enabling remote measuring and control [6]. As a basic function of PV SIs, the Constant Power Factor (CPF) function enables PV generators to inject active and reactive power on a certain power factor [7]. The CPF function is effective in mitigating voltage rises, especially at the end of a feeder. Moreover, it is easy to apply as only one parameter, that is, Power Factor (PF), needs to be set. As local voltage profile various among PV customers, CPF function is limited by the lack of flexibility, leading to occasional over or insufficient reactive power injection as the reactive power output do not refers to the local voltage measurement. Moreover, it has been shown in reference [8] that CPF function tends to increase the tap operations when the PV output is unstable, potentially reducing the lifespan of OLTC.

Different from CPF function, volt-var (VV) function enables PV SIs to adjust reactive power injection according to local voltage measurement, effectively and flexibly solving the voltage regulation problems [9]. However, basic VV function faces difficulties when applied into real power systems. To properly implement the basic VV function in operation, the distribution system operator (DSO) needs to determine the control parameters defining the VV curve. The determination methods have been discussed in many researches [10–20]. Reference [10] showed different VV curve settings under different rules, and tested the control effects on a real distribution system model. References [11–15] showed that in order to apply the VV function into a large-scale model, better

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control effect can be obtained by setting VV curve considering the local voltage profiles. Considering the changes in voltage profiles due to unstable PV generation, Refs [16–18] proposed tuning the VV parameters in real-time according to PV penetration rate, weather conditions or voltage stability. However, advanced data communication and process is necessary for real-time adjustment methods, which is not cost-effective. Therefore, considering practical operation, PV SIs are expected to determine the proper reactive power output dynamically depending on local measurement and process, which has not been studied in detail. Moreover, as voltage fluctuation problem becomes severe on cloudy days because of the unstable PV generation output, momentary voltage violation also harms the distribution system stability, which difficult to be mitigated by OLTC. However, basic VV function is not effective in mitigating voltage fluctuations [21]. Also, overmuch reactive power output from certain PV customers is possible to cause active power curtailment, leading to unfairness among PV customers. As unfairness problem can harm the promotion of PV SIs, the variation of reactive power output among PV customers should be reduced [22].

In this study, a decentralized voltage regulation scheme is proposed, which involves implementing a combination of PID closed-loop based VV function and CPF function within PV SIs. PID controllers are well-known as a control algorithm in industrial applications, and enable accurate and responsive correction output. By introducing PID Controller into the VV function of PV SIs, the parameter setting is simplified as the PID controller automatically calculates the reactive power output based on proportional, integral, and derivative terms, and only a voltage reference needs to be set by DSO. In addition, due to the responsive output correction of PID controller, it tends to mitigate the fast voltage fluctuation better than the basic VV function. Then, to mitigate the unfair reactive power output among PV customers, a mode-switching function based on the proposed PID closed-loop based VV function and CPF function is proposed.

To verify the effectiveness, the proposed PID closed-loop based VV function and mode-switching function are tested on a 6-feeder medium-low voltage distribution simulation system, and evaluated in terms of the local voltage violation, total reactive power output, fairness among customers, and the tap operations of OLTC.

The remainder of this paper is organized as follows. In Section 2, the principle of PID Close-Loop based VV function of PV SIs is explained, then the mode-switching function based on PID closed-loop based VV function and CPF function is described. In Section 3, the simulation setting is introduced. In Section 4, we present and analyze the simulation results. Finally, in Section 5, we summarize the key contributions and findings of this paper.

2. The Proposed PID Closed-Loop Based VV Function and Mode-Switching Function

2.1. Overview

As summarized in Section 1, the basic VV function determines the reactive power output referring to a VV curve shown in Fig. 1. To set the VV curve properly, DSO should estimate eight parameters (V1, V2, V3, V4, Q1, Q2, Q3, Q4) according to conditions such as PV penetration rate, local voltage sensitivity, and so forth. This complicated parameter setting significantly affects the utilization of VV function.

Therefore, a PID closed-loop control algorithm is implemented into the VV function of the PV SIs in this study. As a result, the DSO only needs to set one parameter, voltage reference, and

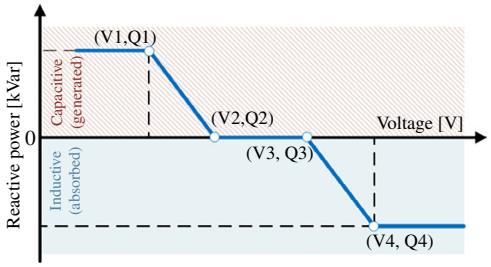


Fig. 1. Basic VV curve

PID controller will calculate the proper reactive power output for certain PV generator. Consequently, the management of a mass of PV SIs in the distribution system is simplified for the DSO. Moreover, compared to basic VV curve, PID controller can mitigate fast voltage fluctuation and handle the momentary voltage violation better.

However, the implementation of the PID closed-loop based VV function leads to a situation where the local voltage profiles are close to voltage violation are prompted to adjust their reactive power output more assertively. Conversely, PV generators with voltage profiles well within the acceptable range do not contribute reactive power. This dynamic gives rise to an issue of unfairness, particularly when certain PV SIs generate an excess of reactive power, which could potentially result in the curtailment of active power generation. To solve the unfairness problem, the proposed mode-switching function based on the proposed PID closed-loop VV function and CPF function is described in Section 2.3.

2.2. PID Closed-Loop based VV function

Figure 2 shows the control diagram of the PV SIs using the proposed PID closed-loop based VV function. First, the maximum AC power P_{max} that can be generated by the PV panel is calculated by a Maximum Power Point Tracking (MPPT) function according to the DC voltage of the PV panel, V_{panel}. Then, the maximum available reactive power output Q_{max} is calculated with regard to the inverter capacity S_{inv}. To prevent undesired generation curtailment of PV generators, priority is given to active power generation in the inverter capacity usage, which means that reactive generation can only use the inverter capacity retained by active power generation according to (1),

$$|Q_{out}| \le |Q_{max}| = \sqrt{S_{inv}^2 - P_{out}^2} \tag{1}$$

At the same time, the voltage at the Point of Common Coupling (PCC), V_{mea} is measured in real-time and locally. Then, V_{mea} and Q_{max} are input into the PID controller. The PID controller with a closed-loop feedback calculates the adequate control value (Q_{out}) to control V_{mea} to the voltage reference (V_{ref}) set by the DSO by processing voltage error $V_{error}(n)$ between V_{mea} and V_{ref} according to (2). Based on the current error $V_{error}(n)$, the error last time $V_{error}(n-1)$, and the error the time before last time $V_{error}(n-2)$, the variation of reactive power output reference ($\Delta Q_{ref}(n)$) from that in the last time $Q_{ref}(n-1)$ is calculated through proportional process, integral process, and derivative process according to (3). K_p, K_i, and K_d denote the coefficients for the proportional, integral, and derivative terms respectively. Summing $\Delta Q_{ref}(n)$ and $Q_{ref}(n-1)$, the current output control value ($Q_{ref}(n)$) is calculated by (4).

$$V_{error}(n) = V_{mea}(n) - V_{ref}(n) \tag{2}$$

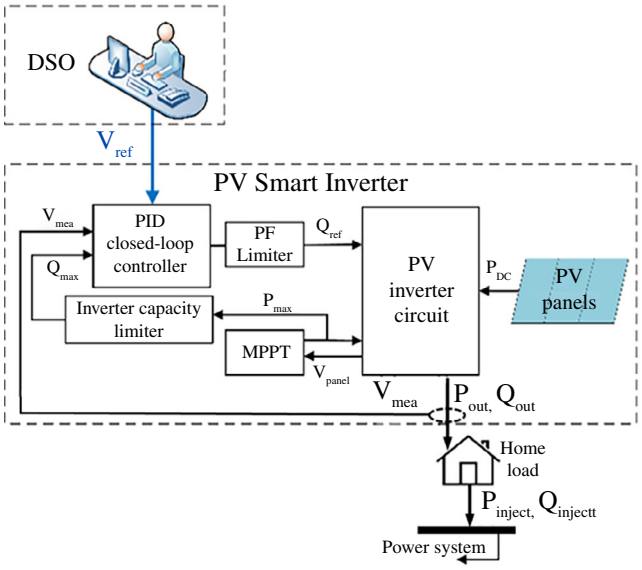


Fig. 2. Block diagram of the PID closed-loop based VV function of PV SIs

$$\Delta Q_{ref}(n) = \overbrace{K_p\left[V_{error}(n) - V_{error}(n-1)\right]}^{proportional} + \overbrace{K_i V_{error}(n)}^{integral} + \overbrace{K_d\left[V_{error}(n) - 2V_{error}(n-1) + V_{error}(n-2)\right]}^{derivative} \tag{3}$$

$$Q_{ref}(n) = Q_{ref}(n-1) + \Delta Q_{ref}(n) \tag{4}$$

Under the control effect proportional term of the PID controller alone, steady-state voltage error will remain, because the proportional controller requires an error to generate an output response. With the integrate term, the integral controller can integrate the past errors and generate output response so that the steady-state error is eliminated. Finally, the derivative term can estimate the future trend by calculating the changing rate of voltage error, strengthening the controller's ability to mitigate fast voltage fluctuations.

Then, according to (5–6), the PF limiter limits the reactive power output so that the Q output can meet the PF standard determined by the DSO to prevent over reactive power injection from PV SIs (Fig. 3).

$$\left|Q_{ref}\right| \le \left|Q_{PFmax}\right| = \frac{\sqrt{1 - PF_{min}^2}}{PF_{min}} P_{out} \tag{5}$$

$$PF_{min} = \cos\left(\theta_{max}\right) \tag{6}$$

where PF_{min} is the minimum Power Factor at which low-voltage (LV) PV generators are allowed to inject power to the power system. PF_{min} is defined in Eq. (6), where θ_{max} represents the maximum power factor angle. Q_{ref} should not exceed the maximum reactive power output limited by the minimum power factor, otherwise it can harm the system stability.

Next, the reference value of reactive power output Q_{ref} is calculated and input into the PV inverter circuit. Neglecting the interior mechanism of inverter circuit, the actual reactive power output Q_{out} of PV SIs can be considered equivalent to the reference value Q_{ref}. Moreover, as the active power generation is prioritized,

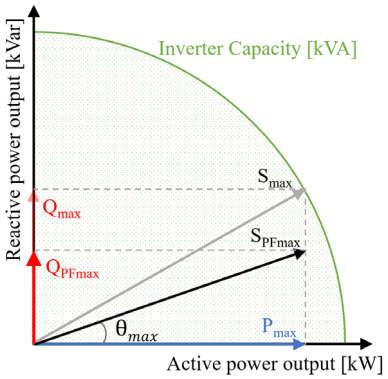


Fig. 3. Schematics showing PF limit effecting the Q output of PV SIs

P_{out} should be equal to P_{max} assuming there is no active power curtailment.

Finally, subtracting the home load from the P_{out} and Q_{out} of PV SIs, the surplus active power P_{inject} and reactive power Q_{inject} are injected into the power system. As a result, once the V_{ref} for the PV SIs in the distribution system is set by the DSO, PV SIs will regulate the voltage automatically so that the local voltage can be maintained as close to V_{ref} as possible and mitigate the voltage fluctuations.

2.3. Mode-switching function

As shown in the control flowchart in Fig. 4, when the measured local voltage V_{mea} is between the upper voltage threshold V_{up} and lower voltage threshold V_{low}, PV SIs will work with the CPF function to guarantee a fairer contribution to voltage regulation. Moreover, when the locally measured voltage, V_{mea} exceeds the upper or lower thresholds, the PV SIs will switch to the PID closed-loop based VV function whose voltage reference V_{ref} is set to V_{up} or V_{low}, in order to avoid voltage violation and mitigate voltage violation better. Moreover, as shown in Fig. 5, to apply the proposed mode-switching function to PV SIs in the power system, the DSO need to set three parameters (V_{up}, V_{low}, PF), which is much more simpler than eight parameters of basic VV function.

In summary, the proposed mode-switching function enables PV SIs to mitigate the voltage fluctuation better, makes the parameter setting simpler, and can prevent the unfairness problem among PV customers.

3. Numerical Simulation Setting

3.1. Simulation model

To examine the control effect of the proposed voltage regulation scheme using the PID closed-loop based VV function and mode-switching function of PV SIs, numerical simulation was conducted on a 6-feeder medium-low voltage distribution system model based on the JST-CREST 126 distribution feeder model, as shown in Figs 6 and 7 [23,24].

The distribution system model comprises six 6.6 kV MV distribution feeders, originating from the distribution substation, and 100 V low-voltage systems are connected to the MV distribution system through pole transformers (6600 V/105 V). The distribution length of Feeder 1 is 4.356 km, Feeder 2 is 5.919 km, Feeder 3 is 4.424 km, Feeder 4 is 3.589 km, Feeder 5 is 3.186 km, and Feeder 6 is 4.727 km. One OLTC is installed in the distribution substation, working for the centralized OLTC control described in

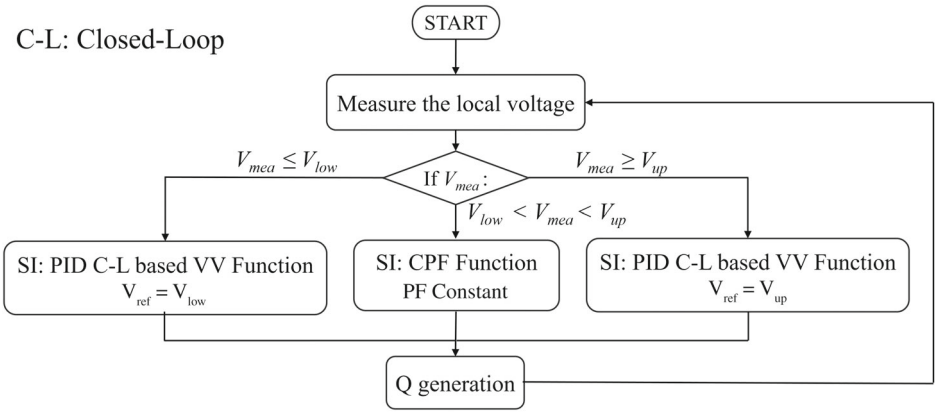


Fig. 4. The flowchart of mode-switching function of PV SIs

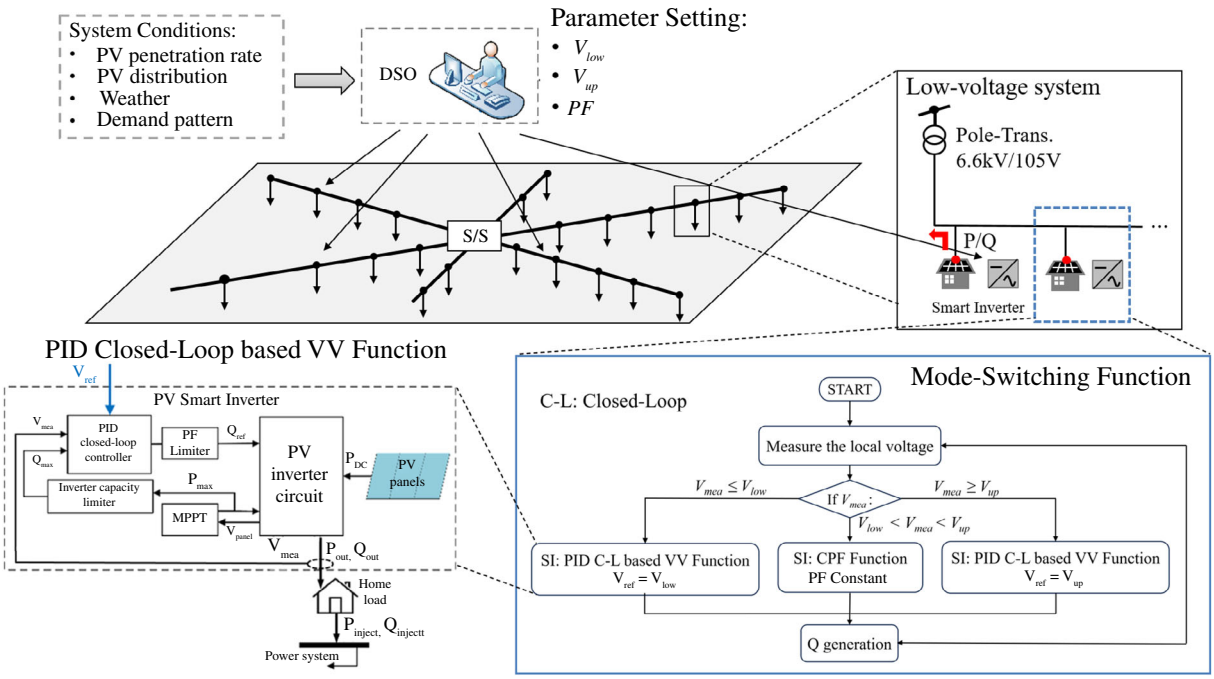


Fig. 5. The overview voltage regulation scheme including PID closed-loop based VV function and mode-switching function of PV SIs

Section 3.2, and all the LV customers are assumed to have installed a PV generator with PV SIs.

Table I shows the detailed parameter setting of the simulation model. The PV penetration rate (PR) in the six feeders was set unevenly by adjusting the capacity of installed PV generators, which makes the local voltage profile of PV customers vary. The PV PR in each feeder $PVPR(F)$, defined by (7), as the ratio of the maximum of total PV generation in one feeder $P_{PV}(t,F)$ in the simulation period and maximum of total demand in one feeder $P_{Demand}(t,F)$ in the simulation period.

$$PVPR(F) = \frac{\max_{t=0}^{T} P_{PV}(t,F)}{\max_{t=0}^{T} P_{Demand}(t,F)} \tag{7}$$

where t represents the simulation time series, and F denotes the feeder number.

The simulation period is for 3 days, including 1 sunny day (day 1) and 2 cloudy days (day 2, 3). Because voltage violation tends to happen on days when solar radiation is strong, that is, sunny days, and voltage fluctuation is strong especially on cloudy days, the simulation period during the described 3 days can well characterize the voltage regulation problem in the distribution system with high PV penetration rate.

3.2. Control scheme of OLTC The OLTC [25] is assumed to collect data measured by sensor switches located in the MV distribution line in each feeder (shown in Fig. 6). Based on the measured data, the OLTC will decide the tap operation in 10-min interval by minimize the objective function $F(t,tap)$ (8), so that the voltage in MV system can be kept in an adequate range (Fig. 8).

The objective function is constructed according to (9). $F(t,tap)$ is the sum of the voltage violation of data measured by every sensor switch in three phases. The voltage violation is defined by (10). The tap position is determined by minimizing the objective

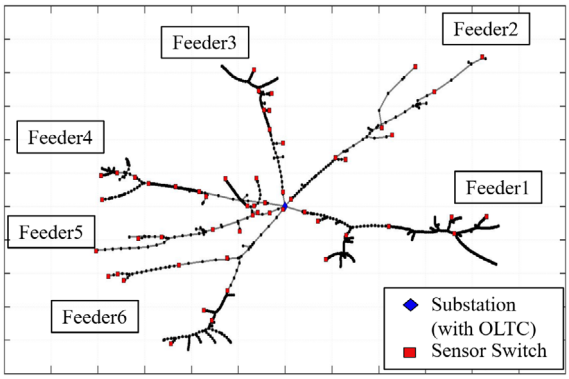


Fig. 6. The MV system of the 6-feeder simulation model

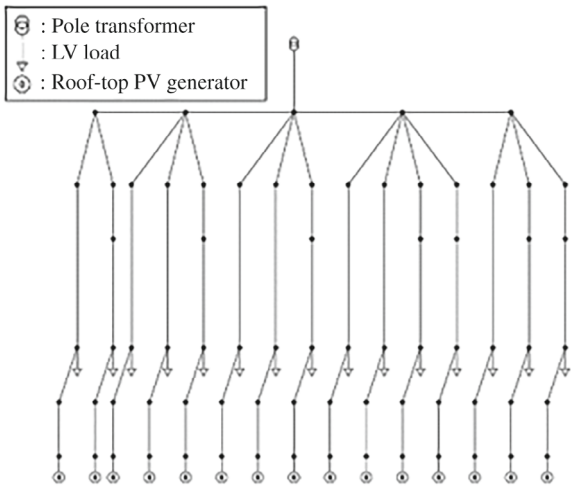


Fig. 7. One example of a LV system

Table I. Parameter setting of the simulation model

MV customers number		112
LV customers number		6057
LV customers' PV number		6057
PV Capacity (PVPR)	Feeder 1	13.6 MW (160%)
	Feeder 2	5.10 MW (77%)
	Feeder 3	5.53 MW (75%)
	Feeder 4	3.50 MW (54%)
	Feeder 5	0.21 MW (4%)
	Feeder 6	0.80 MW (12%)
Simulation period		3 days
Simulation time interval		10 s

function among the three possible operations in the next step according to (8). $(tap_t - 1, tap_t, tap_t + 1)$ represents the tap-down operation, tap-keeping operation, and tap-up operation. The voltage after the next operation is estimated by (11) and (12).

$$\underset{tap=tap_t-1,tap_t,tap_t+1}{minimize} \; F(t,tap) \tag{8}$$

$$F(t,tap) = \sum_{p=1}^{3}\sum_{s=1}^{N}(Vio(t,tap,s,p))^2 \tag{9}$$

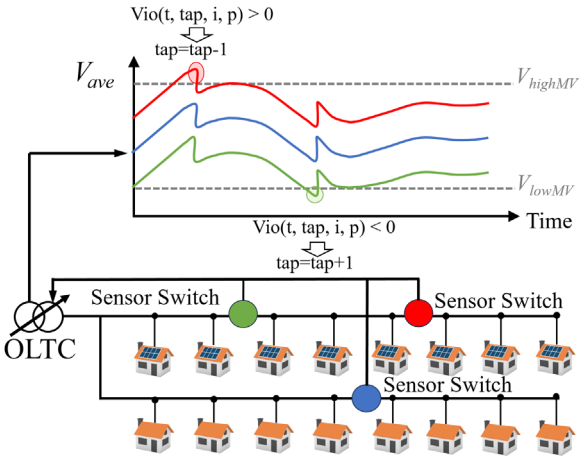


Fig. 8. The tap control scheme of the OLTC

$$Vio(t,tap,s,p) = \begin{cases} V_{ave}(t,tap,s,p) - V_{highMV} & if\ (V_{highMV} < V_{ave}) \\ V_{ave}(t,tap,s,p) - V_{lowMV} & if\ (V_{lowMV} > V_{ave}) \\ 0 & Otherwise \end{cases} \tag{10}$$

$$V_{ave}(t,tap+1,s,p) = V_{ave}(t,tap,s,p) + Tapwidth \tag{11}$$

$$V_{ave}(t,tap-1,s,p) = V_{ave}(t,tap,s,p) - Tapwidth \tag{12}$$

In (8–12), t is the control time series of the OLTC, *tap* is the tap position of the OLTC, s represents the sensor whose data are used by the OLTC, and p represents the phase in the 3-phase system. $V_{ave}(t,tap,s,p)$ is the average measured voltage in the current control period, V_{highMV}, V_{lowMV} are the upper and lower the OLTC control voltage threshold in MV system, *Tapwidth* is the voltage variation from each tap operation.

3.3. Evaluation indexes

3.3.1. Local voltage violation The total local voltage violation, $V_{vioTotal}$ measured at the PCCs of PVs during the simulation period is calculated by (13–14), where t is the control time series of the PV SIs, i represents the PV customer number in the system, V_{uplim} and V_{lowlim} are the upper and lower limits according to the Japanese standard for 105 V low voltage system, which is 95–107 V.

$$V_{vioTotal} = \sum_{i}^{I}\sum_{t}^{T} Vviolation(t,i) \tag{13}$$

$$Vviolation(t,i) = \begin{cases} V_{uplim} - V_{mea}(t,i) & if\ (V_{uplim} < V_{mea}) \\ V_{mea}(t,i) - V_{lowlim} & if\ (V_{lowlim} > V_{mea}) \\ 0 & Otherwise \end{cases} \tag{14}$$

3.3.2. Reactive power output The total reactive power output amount ($Q_{outTotal}$) and reactive power output variation ($Q_{outVari}$) among all customers are evaluated to confirm the regulation efficiency and fairness among customers according to (15–17), where t is the control time series of the PV SIs, and i represents the PV customer number in the system. In (15),

Table II. Cases for validation

Case	PV SI regulation	OLTC regulation
Case 1	CPF function	Centralized tap control
Case 2	Basic VV function	Centralized tap control
Case 3	PID closed-loop based VV function	Centralized tap control
Case 4	Mode-switching function (PID closed-loop based VV function + CPF function)	Centralized tap control

$Q_{outDtotal}$ gives the daily reactive power output amount of each PV customer. Using (16), the total reactive output amount among all the PV customers, $Q_{outTotal}$, is calculated. Then, using (17), the variation of the daily reactive power output amount of each PV customer, $Q_{outVari}$ is calculated to evaluate the fairness, where $\bar{Q} - 0.0001pt_{outDtotal}$ represents the mean value of $Q - 0.0001pt_{outDtotal}$.

$$Q_{outDtotal}(i) = \sum_{t}^{T} Q_{out}(t,i) \tag{15}$$

$$Q_{outTotal} = \sum_{i}^{I} Q_{outDtotal}(i) \tag{16}$$

$$Q_{outVari} = \sum_{i=1}^{I} \frac{\left(Q_{outDtotal}\ (i) - \bar{Q}_{outDtotal}\right)^2}{I} \tag{17}$$

3.3.3. Tap operations of the OLTC The tap operations of the OLTC are evaluated, because over-frequent tap operation can lead to the risk of mechanical wear and contact degradation, potentially deceasing the life span and increasing the maintenance cost. Meanwhile, daily maximum of 27 times of tap operations can be regarded as acceptable, causing no significant impact on the life span of the OLTC [26].

3.4. Cases for validation

To validate the control effect of the proposed method, the PID closed-loop based VV function and mode-switching function is compared to the CPF function and basic VV function (Table II). The four cases are evaluated in terms of the indexes introduced in Section 3.3.

3.5. Parameter setting for PV SI and OLTC

Table III shows the parameter setting for the OLTC centralized control. The upper threshold voltage V_{highMV} and lower threshold voltage V_{lowMV} are adopted referring to Ref. [25], and they are adjusted slightly to ensure that the LV voltage can be maintained in the proper range (95–107 V) in this study.

Table IV shows the parameter setting for the PV SIs. Referring to the practical application cases in Japan, the power factor value of CPF function in Case1 is set to be 0.95 [27]. Also considering the Japanese standard, in Case 2, LV PV generators are not allowed to generate capacitive reactive power, therefore ($V1$, $V2$) of the VV curve is set to be zero. ($V3$, $V4$) in this work are set according to the default values provided in Hawaiian Electric's Source Requirement Document v1.1 [28], and adjusted to fit the simulation system so that the voltage violation in LV system is minimized. Aiming to reduce the voltage violation and fluctuation near upper limit of proper voltage range in Japan, the parameter of proposed PID closed-loop based VV function (V_{ref}) is determined based on experimental results in 103–106 V. Additionally, the parameters of mode-switching function (V_{up}, V_{low}, Constant PF) are determined so that both the upper voltage violation and lower voltage violation are controlled.

Table III. Parameter setting for the OLTC

Meters on sensor switches	55
Control time interval	10 min
Tap width	1.5%
Initial tap position	8
Total tap positions	16
V_{highMV}	1.01 p.u.
V_{lowMV}	0.94 p.u.

Table IV. Parameter setting for the PV SIs

CPF function	PF value	0.95 (inductive)
Basic VV function	PF limit	0.9 (inductive)
	VV3	103.5 (V)
	VV4	106.5 (V)
	Q3	0 (kVar)
	Q4	QmaxPF (Fig. 3)
PID closed-loop based VV function	PF limit	0.9 (inductive)
	K_p	0.15
	K_i	0.1
	K_d	0.001
	V_{ref}	104.5 (V)
Mode-switching function	V_{up}	104.5 (V)
	V_{low}	101 (V)
	Constant PF	0.98 (inductive)

4. Numerical Simulation Result and Discussion

4.1. Local voltage violation

As shown in Fig. 9, compared to Case 1, the total voltage violation in evaluation period is reduced in Case 2, Case 3, and Case 4 by using the basic VV function, PID closed-loop based VV function and mode-switching function of PV SIs. The voltage violation is significant in Case 1 as the CPF function leads to huge voltage fluctuations when PV generation is unstable, making voltage more difficult to control. Then, the voltage violation is reduced in Case 2 by 98.5% by the basic VV function, though voltage violation remains. In Case 3, by using the proposed PID closed-loop based VV function, voltage violation is reduced by 99.4% than that in Case 1 and by 61.4% than that in Case 2. This is because PID closed-loop based VV function can mitigate momentary voltage violation caused by fast voltage fluctuations near the upper limit better than the basic VV function. Moreover, in Case 4, the mode-switching function enables more customers who can afford to attend the voltage regulation, alleviating the reactive power generation burden of certain customers whose local voltage profile is at risk, further reducing the voltage violation by 99.9% than in Case 1 and by 99.0% than in Case 2. That verifies that the proposed mode-switching function improves the performance of PV SI on voltage violation reduction.

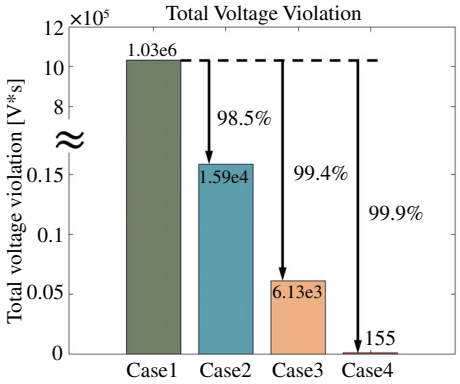


Fig. 9. Total voltage violation of evaluated days

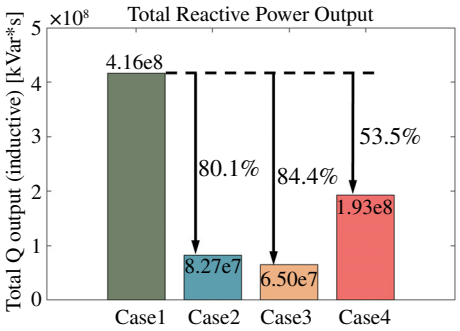


Fig. 10. Total reactive power output of evaluated days

4.2. Total reactive power output and fairness

Figs 10–12 show the evaluation results of total reactive power output and fairness. As shown in Fig. 10, the reactive power generation is reduced by 80.1% in Case 2 compared to that in Case 1. Moreover, the reactive power generation is further reduced in Case 3 by 84.4% compared to that in Case 1. Meanwhile, the reduction of the reactive power generation is less in Case 4 than in Case 2 and Case 3, but it is still 53.5% less than in Case 1. As shown in Fig. 11, the reactive power output variation among PVs is the lowest in Case 1, and it increases by 320% in Case 2, by 475% in Case 3, and by 122% in Case 4. Also, Fig. 12 shows the distribution histogram of the reactive power generation of PVs. It can be found that in Case 2 and Case 3, reactive power outputs are distributed in a wide range. Also, there is large difference on the reactive power generation amount among PV customers. On the other hand, in Case 1 and Case 4, the reactive power generation amount of PV customers concentrates compared to Case 2 and Case 3. Moreover, it can be found that in Case 1, the total reactive power amount is mainly concentrated on the right side of the figure, while in Case 4, the total reactive power is mainly concentrated on the left side of the figure.

In this paper, as the PF value is set the same for all the PV SIs, the variation of the reactive power output among PV customers is small, indicating high fairness as shown in Figs 11 and 12. On the other hand, the CPF function determines the reactive power output referring to PV active power output only, while there are PVs whose local voltage does not need to be regulated. That leads to unnecessarily large reactive power output amount as shown in Fig. 10.

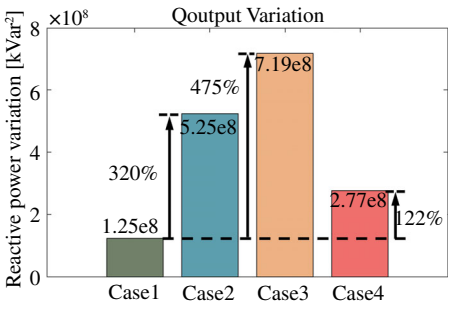


Fig. 11. Reactive power output variation among PVs

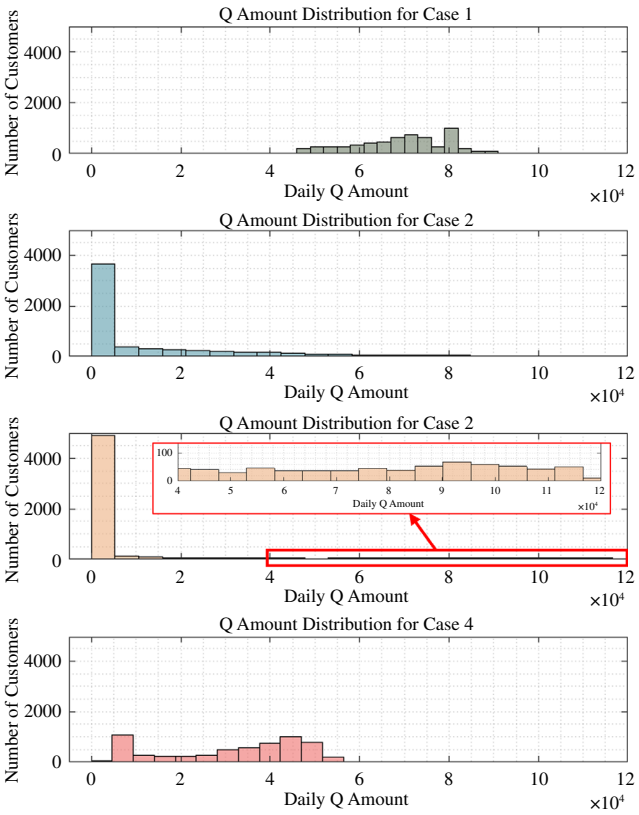


Fig. 12. Reactive power output distribution among PVs

In Case 2, the basic VV function reduces the reactive power generation compared to the CPF function as shown in Fig. 10. However, the reactive power output variation increases among customers compared to the CPF function according to Fig. 11. Moreover, in Case 3, the proposed PID closed-loop based VV function reduces the total reactive power output further compared to the basic VV function as shown in Fig. 10, because the PID closed-loop based VV function allows PV customers whose local voltage profiles are at risk of voltage violation to regulate voltage more aggressively and efficiently and allows PV customers whose local voltage profiles are in the adequate range to output no reactive power. However, the reactive power output variation increases among customers compared to the CPF function and the basic VV function as shown in Figs 11 and 12. This is because in Case 3, V_{ref} is set the same in the whole system, and PV customers whose voltage profiles are in an adequate range output no reactive

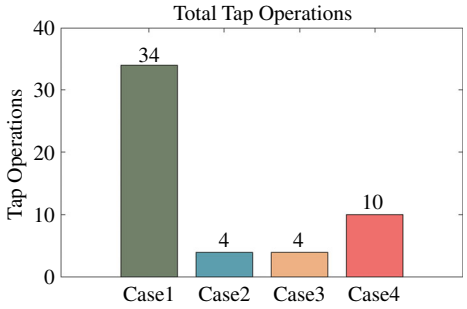


Fig. 13. Total tap operations of the OLTC

power according to the control logic of the PID closed-loop based VV function, affecting the fairness.

As for the proposed mode-switch function, the variation among customers is reduced compared to the basic VV function and the PID closed-loop based VV function, but still being larger than the CPF function according to Figs 11 and 12. However, the total reactive power output increases compared to the basic VV function and the PID closed-loop based VV function, but still 53.5% less compared to the CPF function according to Fig. 10, as the reactive power generation of the customers located at the beginning of a feeder is not as effective on voltage regulation as the customers located at the end of a feeder because of the impedance variation.

4.3. Tap operations of the OLTC As shown in Fig. 13, the tap operations of OLTC occur 34 times during the evaluated 3 days in Case 1. That is because that CPF function can lead to more voltage fluctuations, and OLTC has to operate more frequently to keep the voltage in an acceptable range. Meanwhile, both in Case 2 and Case 3, the tap operations are four times, reduced remarkably than in Case 1. However, more tap operations can be found in Case 4 than Case 2 and Case 3 because of the additional voltage drop when partial PV SIs are switched to CPF mode. As the additional tap operations result in better voltage regulation effect, as described in Section 4.1, the additional tap operations are thought to be necessary as it will not cause significant impact on the lifespan of the OLTC as described in Section 3.3.

4.4. Detailed discussion on day 3 (cloudy) Figures 14–17 illustrate the waveforms of the maximum and minimum local voltage measured at the PCCs in each feeder, as well as the total reactive power generated by PVs in each feeder and the tap operations of the OLTC in day 3.

On day 3, the weather conditions were characterized by occasional cloud cover and intense solar radiation. As a result, this day witnessed significant challenges related to voltage violations and fluctuations. Therefore, the effectiveness of the proposed control method became apparent when examining the voltage waveforms.

In particular, as shown in Fig. 14, the CPF function leads to wild voltage fluctuations, and momentary voltage violations as well. In addition, the OLTC tap operation occurs frequently to compensate for the over-large voltage fluctuations caused by the CPF function. In addition, it can be found that the PV SIs generate a great amount of reactive power.

Meanwhile, when using the basic VV function, although some of the PV generators (Feeder 1, Feeder 2) have generated great

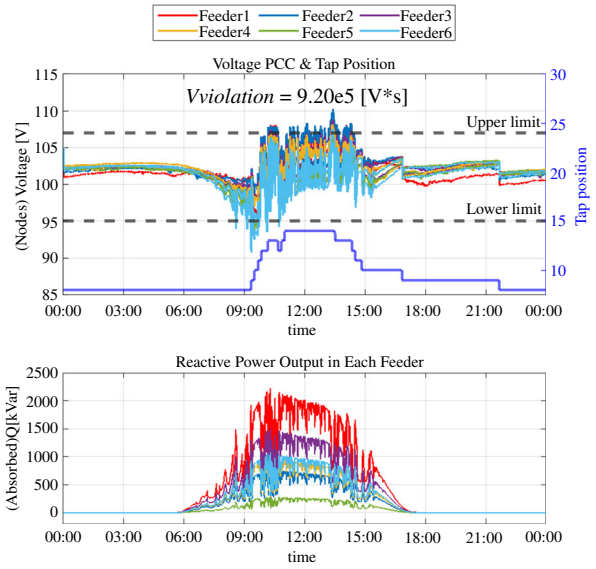


Fig. 14. Voltage, tap position, reactive power output waveform of Case 1 on day 3

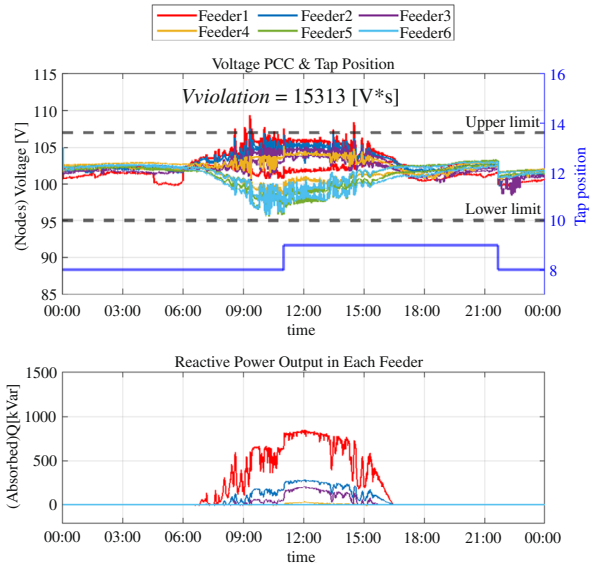


Fig. 15. Voltage, tap position, reactive power output waveform of Case 2 on day 3

amount of reactive power, the basic VV function is unable to mitigate fast voltage fluctuations, leading to momentary voltage violation as shown in Fig. 15.

In Fig. 16, by the implementing PID closed-loop based VV function, it is evident that the voltage fluctuations near the upper limit is reduced, leading to less momentary voltage violation. However, it can be found that there are a large proportion of PV customers whose PVs do not generate any reactive power, while the PV customers whose local voltage is about to violate generate great amount of reactive power. That causes unfairness problem.

In Fig. 17, the mode-switching function draws to a more favorable voltage regulation result by letting PV customers whose local voltage profiles are in an adequate range generate reactive

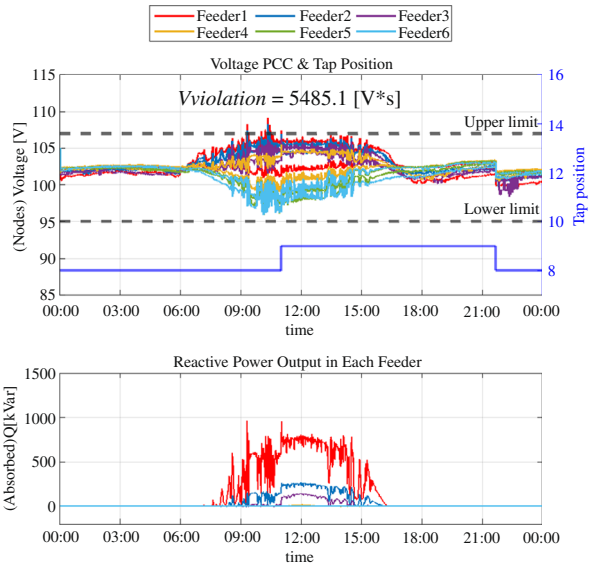


Fig. 16. Voltage, tap position, reactive power output waveform of Case 3 on day 3

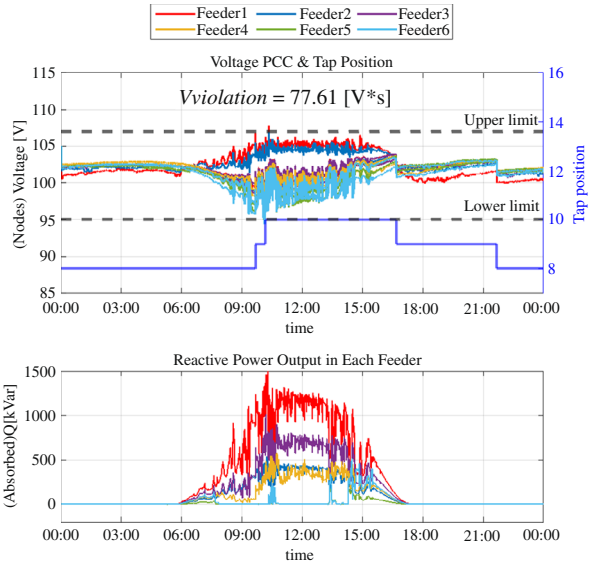


Fig. 17. Voltage, tap position, reactive power output waveform of Case 4 on day 3

power at constant PF. This reduces the burden on certain PV customers, such as those in Feeder 1 and Feeder according to Fig. 17, while additional voltage drop occurs compared to Case 1 and Case 2. This additional voltage drop can be resolved by the OLTC tap control, therefore additional OLTC tap operations can be observed in Case 3 compared to that in Case 1 and Case 2. In addition, from the waveform of reactive power generation, it can be seen that more PV customers attend to the voltage regulation by generating reactive power, mitigating the unfairness problem.

The significance of these findings highlights the effectiveness of the proposed control approach to address voltage stability concerns, especially during challenging conditions such as those encountered on day 3.

5. Conclusion

In this study, we discuss new challenges in voltage regulation in distribution systems with a high installation rate of distributed PV generators. To address the problem of voltage fluctuation and momentary voltage violation, we propose a novel decentralized control method called the PID closed-loop based VV function of PV SIs. This function simplifies the parameter setting of basic VV functions based on a VV curve and effectively mitigates fast voltage fluctuations to prevent momentary voltage violations. Furthermore, to reduce the unfairness problem among PV customers, we introduce a mode-switching function based on the PID closed-loop based VV function and CPF function. Lastly, we test the effectiveness of the proposed control scheme of PV SIs on a 6-feeder distribution system simulation model with uneven PV penetration rates over 3 days, featuring a mix of sunny and cloudy weather conditions. The results demonstrate that the proposed PID closed-loop based VV function outperforms the CPF function and basic VV functions of PV SIs in reducing voltage violations by 99.4%. Moreover, the mode-switching function succeeded to reducing the voltage violations by 99.9%, almost eliminating the voltage violation. Moreover, it is observed that the PID closed-loop based VV function tends to reduce reactive power output by 84.4%, and leads to a variation increase of 475% than these in the CPF function for reactive power generation among PV customers, potentially affecting fairness. To address this, the model-switch function is introduced, significantly reducing the variation with only 122% larger than that of the CPF function, also reducing the total reactive power output by 53.5% than that of the CPF function.

In a nutshell, we identify the effectiveness of the novel decentralized voltage regulation scheme of PID closed-loop VV function and mode-switching function of PV SIs, which is expected to be widely utilized in future distribution systems. Nevertheless, we also note a trade-off relationship between the total reactive power output of VV function and fairness among customers, because the efficiency of reactive power voltage control method varies with the location of PVs in the distribution lines. To develop a voltage control scheme in distribution systems considering both high fairness and efficiency, cooperative control methods should be considered above the base of this study.

In the future studies, we will need to consider the effect of the parameter settings on the control performance. In this study, the parameters are set through trial and error tailored to the system, but assuming actual operation, the proposed method would be required to apply the proposed method into various distribution systems. In this case, from the perspective of the burden on system operators, it is necessary to consider the need for parameter tuning to match the increase in the amount of PV installed and demand changes. We also plan to evaluate the usefulness of these parameters by considering not only changes in the amount of PV installed, but also changes in trends due to weather and demand seasonality.

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